Predicting Tropospheric Ozone

Replicating "CS229 Autumn 2019: Predicting Tropospheric Ozone" (M. Sutton)

UE24CS352A Machine Learning · Mini-Project

Problem

- Ground-level ozone is a pollutant formed from NO_X + VOCs + sunlight
- EPA flags regions above the limit as "nonattainment", which triggers costly regulation
- Goal: predict a site's daily ozone (ppm) from **temperature, pressure, wind, month, location, NO_2, VOC**
- Regression problem, evaluated with R^2 like the paper

Dataset

- EPA Historical Air Quality (same data as the paper's BigQuery tables), 1980–2017
- 6 daily files per year: ozone, NO_2, VOCs, temperature, pressure, wind → ~600 MB
- Average per (site, day), then inner join → **90,997 rows, 86 sites**
- Time split: train 1980–2016 (86,461) · val Jan–Jun 2017 (2,384) · test Jul–Dec 2017 (2,152)
- Min-max scaling fitted on training data only

Pipeline

```
download.py  →  build_dataset.py  →  train.py  →  evaluate.py  →  predict.py
 EPA zips        clean + join         5 models      plots +          demo
                 data/ozone.csv       metrics.csv   importance
```

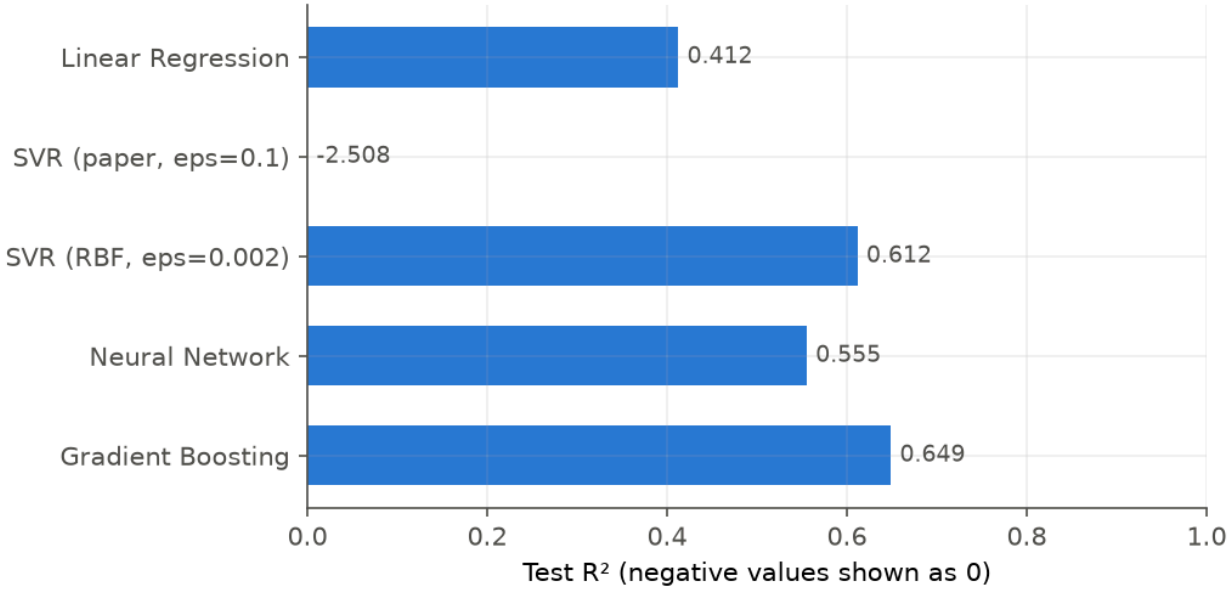
- Python, pandas, scikit-learn, matplotlib
- Whole pipeline runs on a laptop in a few minutes after download

Models

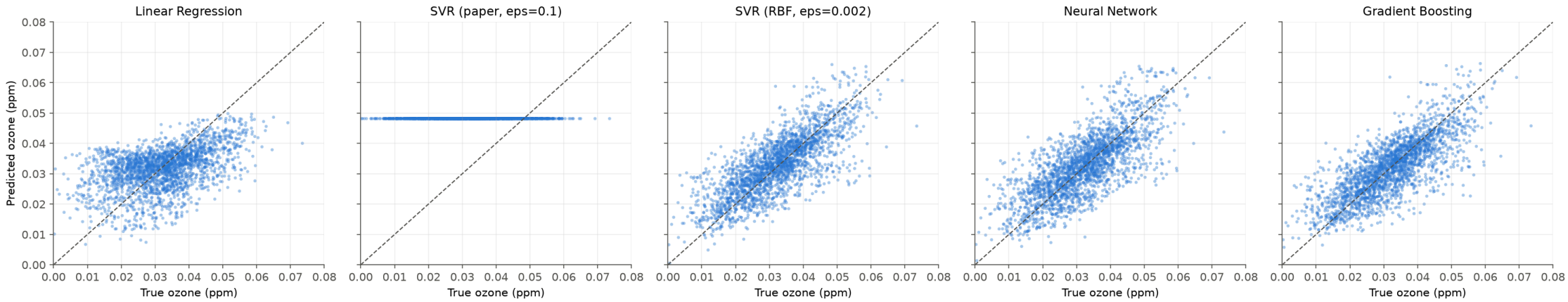
- **Linear Regression**: baseline
- **SVR (paper)**: polynomial kernel, ε = 0.1
- **SVR (fixed)**: RBF kernel, ε = 0.002, because ozone is ~0.03 ppm so ε = 0.1 ignores all errors
- **Neural Network**: 3 × 200 ReLU, adaptive learning rate (paper's best)
- **Gradient Boosting** (our addition): fast, strong on tabular data

Results: test R^2 (Jul–Dec 2017)

Model	Paper	Ours
Linear Regression	0.467	0.412
SVR (ε=0.1)	−2.363	−2.508
SVR (ε=0.002)	–	0.612
Neural Network	0.654	0.555
Gradient Boosting	–	**0.649**

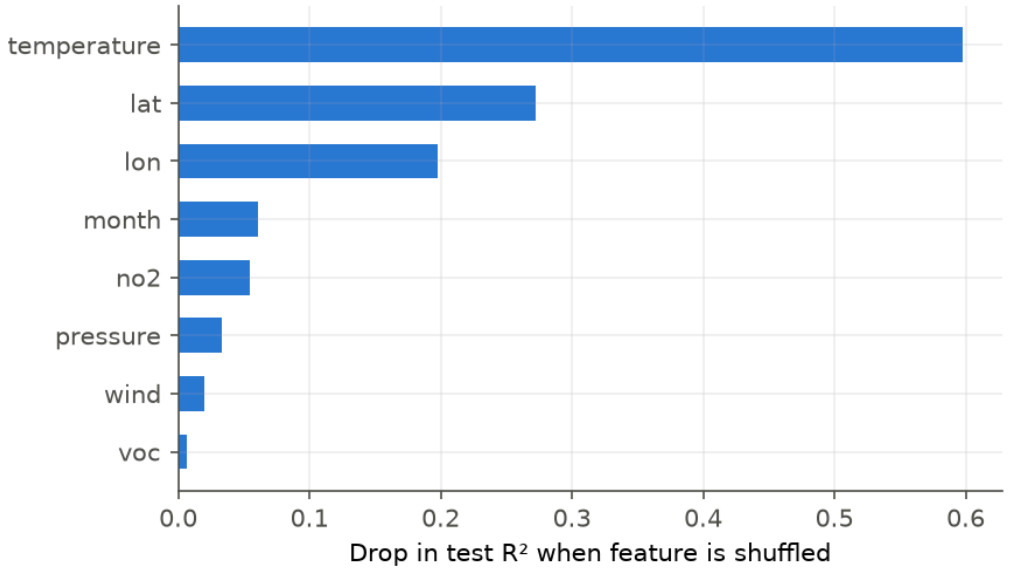


True vs predicted (validation set)



The paper-setting SVR predicts a constant (flat line), which is why its R^2 is negative.

Which features matter?



- Temperature dominates (proxy for sunlight)
- Location next: sites differ a lot
- NO_2 and VOC add less once weather and location are known

Conclusions

- Paper's findings replicate: linear underfits, ε = 0.1 SVR fails, non-linear models reach $R^2 \approx 0.6$–0.65
- Fixing SVR's ε or using gradient boosting matches or beats the paper's neural network, with far less training time
- Future work: previous-day ozone, traffic/industry data, predicting the EPA ozone class directly

Live demo

```
python predict.py --temperature 85 --pressure 1010 --wind 4 --month 7 \
    --lat 34.05 --lon -118.24 --no2 15 --voc 60
```

Thank you. Questions?